



CompoundIQ

AI-Powered Safe 3-Drug Combination Discovery

ITAI 2376 | Deep Learning Capstone | Spring 2026

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The Story Behind CompoundIQ

Patients with complex conditions often rely on combinations of drugs, but the more drugs that are added, the greater the risk of harmful interactions. Since the number of possible three-drug combinations is extremely large, testing every option in the lab is not realistic. CompoundIQ was designed to help narrow that search by identifying promising combinations before costly and time-intensive lab testing begins.

CompoundIQ is our proposed AI system for designing and evaluating three-drug combinations. It takes a therapeutic mechanism written in plain English, generates candidate compounds, predicts both pairwise and three-way interactions, filters out unsafe options, creates new drugs, and then ranks the safest and most promising results. In our project design, this brings together natural language processing, molecular generation, and graph-based interaction modeling into one end-to-end pipeline trained on biomedical and drug-interaction datasets such as ChEMBL, DrugBank, SIDER, and related sources.

End-to-End Architecture



Ingestion

ChEMBL: ~35,000 drug-like molecules
DrugBank: drug interaction records
PubMed: biomedical literature corpus

Data Processing

Handle nulls (mean/median/mode)
Remove anomalies and duplicates
Normalize molecular descriptors

Feature Extraction

Remove low-variance features
Compute pairwise correlations
Identify output-influencing features
(CYP enzymes, centrality, similarity)

Why JT-VAE?

Junction Tree Variational Autoencoder for Molecular Graph Generation

How JT-VAE Works

Unlike other molecular generators that build molecules atom by atom (risking invalid structures), JT-VAE uses a two-phase approach:

- 1 A molecule is decomposed into a junction tree of valid chemical substructures (fragments) extracted from training data
- 2 The tree and molecular graph are encoded into separate latent vectors (z_T and z_G)
- 3 To generate, the model first reconstructs the junction tree, then assembles the fragments back into a complete, valid molecule

This structure-by-structure approach guarantees every generated molecule is chemically valid

Why It Matters: Validity Comparison

Method	Reconstruction	Validity
CVAE	44.6%	0.7%
GVAE	53.7%	7.2%
SD-VAE	76.2%	43.5%
GraphVAE	—	13.5%
Atom-by-Atom LSTM	—	89.2%
JT-VAE (Ours)	76.7%	100.0%

100%

molecular validity — every molecule JT-VAE generates is chemically valid by construction

How the User Interacts



Key: The user never handles raw data. They describe a therapeutic goal in plain English, and the pipeline handles molecular generation, interaction analysis, and safety ranking automatically. Results are displayed in a Streamlit dashboard with an interactive 3D molecular viewer.

Libraries & Tools

Library	Role in CompoundIQ	Component
torch-geometric	Graph neural network layers and message-passing for drug interaction graphs	Pairwise & 3-Way GNN
RDKit	SMILES parsing, molecular validation, junction tree decomposition into fragments	JT-VAE
NetworkX	Building drug interaction graphs and junction tree structures	JT-VAE, GNN
Transformers (HF)	Loading and running BioBERT (dmis-lab/biobert-base-cased-v1.1)	BioBERT
DeepChem	Molecular dataset loading (Tox21, ESOL, BBBP) and featurization utilities	Data Collection
chembl-client	Python API client for querying ChEMBL molecular database directly	Data Collection
Streamlit	Interactive web dashboard for user input and results display	Frontend
Py3Dmol	3D molecular structure rendering embedded in the Streamlit UI	Frontend



CompoundIQ

ANY QUESTIONS?

5,344,397 trainable parameters

5-stage end-to-end pipeline

Novel molecular generation via
JT-VAE